✦ COMPLETE REVISION GUIDE

IB DP Digital Society

Full Syllabus · HL + SL · Score a 7



Digital Systems · Data & Information · Networks · Social Impacts · Ethics & Governance

HL Extensions · Exam Strategy · 20 Pages

IB Digital Society · Based on current syllabus standards · Includes HL extensions · Exam-focused revision

📖 How to Use + Syllabus Map

🗺️ HOW TO USE THIS GUIDE

- **Read sequentially** for full understanding
- **Use Exam Tips** 💡 to focus on high-value content
- **Study Eval boxes** ⚖️ — these are core to IB marking
- **Flag** ⚠️ boxes — common student errors to avoid
- **HL students:** cover all pages including red HL sections
- **Night before:** use Page 20 checklist only
- **Command terms** on Page 18 — memorise before the exam

> 💡 **EXAM TIP** IB Digital Society rewards students who can **evaluate** from multiple perspectives. Never just describe — always analyse and judge.

📋 SYLLABUS TOPICS

S1	Digital Systems & Technologies	pp. 3-5
S2	Data & Information	pp. 6-7
S3	Networks & Connectivity	pp. 8-9
S4	Social Impacts of Technology	pp. 10-12
S5	Governance & Ethics	pp. 13-14
HL	HL Extensions	pp. 15-16
EX	Exam Mastery (Mistakes, Strategy, Terms)	pp. 17-20

🗺️ VISUAL SYLLABUS ROADMAP



Each section builds on the previous — technology creates data, data flows through networks, networks affect society, society requires governance

📝 Paper 1

- Short-answer questions
- Data-based questions
- Command terms: define, describe, outline
- Both SL & HL

📝 Paper 2

- Extended-response questions
- Case study analysis
- Command terms: discuss, evaluate, examine
- Both SL & HL

📐 Internal Assessment

- Digital artefact creation
- Exploration of a digital issue
- HL: extended investigation
- 25% of final grade

🖥️ S1 — Digital Systems: Core Concepts

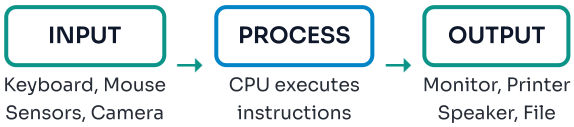
⚙ HARDWARE VS SOFTWARE

Aspect	Hardware	Software
Definition	Physical components	Instructions/programs
Examples	CPU, RAM, HDD, GPU	OS, apps, firmware
Tangible?	Yes	No
Types	Input, Output, Processing, Storage	System, Application, Utility

📦 TYPES OF SOFTWARE

- **System software:** OS (Windows, macOS, Linux) — manages hardware resources
- **Application software:** Word processors, browsers, games — user tasks
- **Utility software:** Antivirus, file managers — system maintenance
- **Open-source:** Code freely available (Linux, Firefox)
- **Proprietary:** Code restricted (Microsoft Office, iOS)

🔄 INPUT–PROCESS–OUTPUT MODEL



+ STORAGE: persistent data saved (RAM = temporary, HDD/SSD = permanent)

🧠 CPU COMPONENTS

- **ALU (Arithmetic Logic Unit):** performs calculations and logic
- **Control Unit (CU):** directs operations, fetch-decode-execute
- **Registers:** ultra-fast temporary storage in CPU
- **Cache:** fast memory between CPU and RAM (L1, L2, L3)
- **Clock speed:** GHz — cycles per second processed

🖥️ DIGITAL SYSTEMS OVERVIEW

🔢 Binary & Data Representation

- All data = binary (0s and 1s)
- Bit → Byte → KB → MB → GB → TB
- ASCII & Unicode for text
- RGB for images (8-bit per channel)
- Sampling rate for audio

⚡ Operating Systems

- Interface between user & hardware
- Functions: memory mgmt, process scheduling, file mgmt, security
- Types: single-user, multi-user, real-time, embedded
- Examples: Windows, Linux, Android, iOS

📱 Embedded Systems

- Dedicated computer inside a device
- Examples: washing machine, pacemaker, car ECU
- Characteristics: real-time, low power, reliable
- Cannot easily be reprogrammed by user

💡 **EXAM TIP** When asked about digital systems, always use specific examples (e.g., name a real OS, real hardware component) and refer to the input–process–output model where relevant.

⚠ **COMMON MISTAKE** Do NOT confuse **RAM** (volatile, temporary) with **ROM** (non-volatile, permanent). RAM loses data on power-off; ROM does not.

🖥️ S1 — Key Technologies: AI · Cloud · IoT

🤖 ARTIFICIAL INTELLIGENCE (AI)

Definition

- Systems that simulate human intelligence to perform tasks
- **Narrow AI:** one specific task (Chess, face recognition)
- **General AI (AGI):** theoretical — human-level across all tasks

🧠 Machine Learning (ML)

- Algorithms learn from data without explicit programming
- **Supervised:** labelled training data (spam filter)
- **Unsupervised:** finds patterns alone (clustering)
- **Reinforcement:** learns via reward/penalty (game AI)

🔗 Deep Learning & Neural Networks

- Layers of artificial neurons
- Powers: ChatGPT, image recognition, voice assistants
- Requires massive datasets

☁️ CLOUD COMPUTING

Definition

- Delivery of computing services over the internet
- No local hardware required

📦 Service Models

- **IaaS:** Infrastructure (AWS EC2) — raw servers
- **PaaS:** Platform (Heroku) — development tools
- **SaaS:** Software (Gmail, Dropbox) — end apps

☁️ Deployment Models

- **Public:** shared (AWS, Azure, Google Cloud)
- **Private:** one org, higher security
- **Hybrid:** mix of both

⚖️ EVALUATION **Cloud Pros:** scalable, accessible, cost-efficient, auto-updates
Cloud Cons: data privacy risks, vendor lock-in, internet dependency, outage risk

📡 INTERNET OF THINGS (IOT)

Definition

- Network of physical devices with sensors, actuators, software connected to the internet

🔧 Key Components

- **Sensors:** collect data (temperature, motion)
- **Connectivity:** WiFi, Bluetooth, 5G, LPWAN
- **Cloud:** processes and stores data
- **Actuators:** take physical action
- **UI:** app or dashboard

🏠 Real-World Examples

- Smart home (Nest thermostat)
- Wearables (Apple Watch)
- Smart agriculture (soil sensors)
- Industrial IoT / Industry 4.0
- Smart cities (traffic sensors)

⚖️ EVALUATION **IoT Pros:** efficiency, automation, data insights
IoT Cons: security vulnerabilities, privacy, data overload, interoperability

💡 EXAM TIP AI, IoT, and cloud computing are frequently examined together. Link them: **IoT devices → generate data → stored in cloud → analysed by AI**. This chain demonstrates systems thinking.

🖥️ S1 — Real-World Applications & Evaluation

🌍 REAL-WORLD TECHNOLOGY USE CASES

Technology	Application	Real Example	Benefit	Limitation
AI	Medical diagnosis	DeepMind / AlphaFold	Faster, more accurate detection	Algorithmic bias, liability
AI	Autonomous vehicles	Tesla Autopilot, Waymo	Safer roads, reduced fatigue	Ethical dilemmas, hacking risk
Cloud	File storage & collaboration	Google Drive, OneDrive	Accessibility, version control	Privacy concerns, outages
Cloud	Streaming	Netflix, Spotify	On-demand, cost-effective	Bandwidth requirements
IoT	Healthcare monitoring	Remote patient monitoring devices	Continuous data, early alerts	Data security, device failure
IoT	Smart agriculture	Soil/weather sensors	Precision farming, less waste	High setup cost, rural connectivity

📊 COMPARING TECHNOLOGIES

Convergence of AI + IoT + Cloud

📡 **IoT Sensor** → Raw data collected

☁️ **Cloud** → Data stored & processed

🤖 **AI** → Patterns detected, decisions made

👤 **User** → Actionable insights delivered

⚠ **COMMON MISTAKE** Never describe technologies in isolation. Examiners reward students who show how technologies **interact** and have **cascading impacts**.

⚖ EVALUATION: AI IN DECISION-MAKING

⚖ **EVALUATION** **Perspective 1 (Optimistic):** AI removes human bias in hiring/lending; processes vast data beyond human capacity; frees humans from repetitive work

Perspective 2 (Critical): AI encodes existing bias from training data; lacks moral judgement; erodes human autonomy; accountability gap ("black box")

Balanced Conclusion: AI is a tool — its impact depends on design, oversight, and governance frameworks

💡 **EXAM TIP** When evaluating, always present **at least two perspectives** and reach a **justified conclusion**. Never just list pros and cons — synthesise them.

🏥 CASE STUDY: AI IN HEALTHCARE

Context

- AI systems analyse X-rays, MRI scans
- IBM Watson, Google DeepMind used in diagnostics
- Deployed in UK NHS trials

Benefits

- Detects cancer earlier than humans
- Reduces radiologist workload
- Available 24/7, no fatigue
- Consistent, standardised analysis

Issues

- Trained on biased datasets (under-represents minorities)
- Liability unclear if AI misdiagnoses
- Patient data privacy concerns
- Over-reliance risk

S2 — Data & Information: Foundations

DATA VS INFORMATION

	Data	Information
Definition	Raw, unprocessed facts	Processed, meaningful data
Example	25, 37, 18	"Average age is 27"
Context	None	Has context & purpose
Value	Low alone	High — enables decisions

DATA COLLECTION METHODS

- **Surveys/Questionnaires:** direct user input; risk of self-selection bias
- **Sensors/IoT:** automatic, continuous; depends on calibration accuracy
- **Web scraping:** automated extraction from websites
- **APIs:** structured data from platforms (Twitter API, Google Maps)
- **Transactional:** e-commerce, banking records
- **Social media:** likes, posts, engagement data
- **Cookies/Tracking:** browser behaviour (raises consent issues)

⚠ **COMMON MISTAKE** Collecting more data ≠ better results. Quality, relevance, and bias matter more than quantity.

BIG DATA

The 5 V's of Big Data

- **Volume:** massive scale (petabytes)
- **Velocity:** generated at high speed (real-time)
- **Variety:** structured, unstructured, semi-structured
- **Veracity:** accuracy and trustworthiness
- **Value:** usefulness for decision-making

Big Data Applications

- **Healthcare:** genomics, epidemic tracking (COVID-19)
- **Finance:** fraud detection, algorithmic trading
- **Marketing:** targeted advertising, customer segmentation
- **Transport:** route optimisation (Google Maps)
- **Government:** census data, policy modelling

⚖ **EVALUATION** **Pro:** enables discoveries impossible by hand; powers personalisation
Con: surveillance risk; data monopolies; privacy erosion; environmental cost of data centres

DATA TYPES

Structured Data

- Organised in rows/columns
- SQL databases, spreadsheets
- Easiest to analyse
- Example: sales database

Unstructured Data

- No predefined format
- Text, images, video, audio
- 80% of all data is unstructured
- Example: social media posts

Semi-Structured Data

- Has some organisation
- JSON, XML, HTML files
- Tags/markers indicate structure
- Example: email with headers

💡 **EXAM TIP** Distinguish between **primary data** (collected first-hand) and **secondary data** (collected by others). Discuss reliability and validity differences in exam answers.

S2 — Data Analysis, Visualisation & Bias

DATA ANALYSIS METHODS

Statistical Analysis

- **Descriptive:** mean, median, mode, range — summarises data
- **Inferential:** draws conclusions beyond the sample (p-value, confidence interval)
- **Correlation:** relationship between variables (≠ causation!)
- **Regression:** predicts outcomes from variables

Machine Learning Analysis

- **Classification:** categorise data (spam/not spam)
- **Clustering:** group similar items (customer segments)
- **Anomaly detection:** find outliers (fraud)
- **Natural Language Processing (NLP):** text analysis

⚠ COMMON MISTAKE **Correlation ≠ Causation**. E.g., ice cream sales and drowning rates are correlated (summer), but one does not cause the other.

DATA VISUALISATION

Types & When to Use

Chart Type	Best For
Bar chart	Comparing categories
Line graph	Trends over time
Pie chart	Parts of a whole
Scatter plot	Correlation between variables
Heat map	Density / geographic data
Histogram	Distribution of data

⚖ EVALUATION **Good visualisation:** clear labels, appropriate chart type, honest scales
Misleading visualisation: truncated axes, cherry-picked data, misleading colour

⚠ BIAS IN DATA

Sampling Bias

- Sample doesn't represent population
- Example: surveying only social media users about internet access
- Fix: random, stratified sampling

Algorithmic Bias

- AI trained on biased data reproduces bias
- Example: COMPAS recidivism tool — racially biased predictions
- Amazon hiring AI rejected women
- Fix: diverse training data, regular auditing

Confirmation Bias

- Analyst interprets data to confirm pre-existing beliefs
- Example: researcher dismisses contradicting data
- Fix: peer review, blind analysis

⚖ EVALUATION **Ethical Dimension of Data Bias:** Biased data systems can perpetuate systemic discrimination. When used in hiring, lending, policing, or healthcare, they affect real lives. Accountability must include: transparency of training data, algorithmic auditing, and mechanisms for appeal.

💡 EXAM TIP Exam questions often present a dataset or scenario and ask you to **identify bias**, **explain its impact**, and **suggest remedies**. Always address all three parts.

S3 — Networks & Internet Structure

TYPES OF NETWORKS

Type	Range	Example
PAN	<10m	Bluetooth, body sensors
LAN	Building	Home/school WiFi
MAN	City	City Wi-Fi networks
WAN	Country/Global	The Internet, leased lines
VPN	Any (encrypted)	Remote work connections

NETWORK TOPOLOGIES

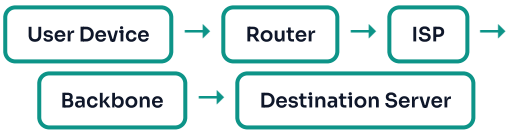
- **Star:** all nodes → central switch; reliable, single point of failure at switch
- **Bus:** all nodes on one cable; simple, entire network fails if cable fails
- **Ring:** nodes in circle; equal access, one failure breaks chain
- **Mesh:** every node connected; highly resilient, expensive
- **Hybrid:** combination; modern corporate networks

INTERNET STRUCTURE

Key Components

- **ISP (Internet Service Provider):** provides internet access (Airtel, BT)
- **Backbone:** high-speed fibre optic cables connecting continents
- **DNS (Domain Name System):** translates domain names → IP addresses
- **IP Address:** unique numerical identifier for every device (IPv4 vs IPv6)
- **Router:** directs data packets between networks
- **Switch:** connects devices within same network
- **Firewall:** filters traffic based on rules

DATA PACKET JOURNEY



PROTOCOLS

TCP/IP

- Foundation of internet
- **TCP:** reliable transmission, checks packets
- **IP:** addressing and routing
- Breaks data into packets; reassembles at destination

HTTP/HTTPS

- Web communication protocol
- **HTTP:** unencrypted (insecure)
- **HTTPS:** encrypted with TLS/SSL
- HTTPS is essential for e-commerce, login pages

Other Key Protocols

- **FTP:** file transfer
- **SMTP/IMAP/POP3:** email sending/receiving
- **SSH:** secure remote access
- **DHCP:** auto-assigns IP addresses

World Wide Web vs Internet

- **Internet:** global network infrastructure (hardware, cables, routers)
- **WWW:** a service that runs on the internet — web pages accessed via HTTP
- WWW ≠ Internet. Email, VoIP, FTP also run on the internet

EXAM TIP A common exam question: "Explain the role of DNS." Answer must include: DNS translates human-readable domain names (google.com) to IP addresses (142.250.x.x), enabling devices to locate servers without memorising numbers.

S3 — Cybersecurity & Data Privacy

CYBER THREATS

Threat	Description	Example
Malware	Malicious software	Ransomware (WannaCry)
Phishing	Fake messages to steal credentials	Fake bank emails
DDoS	Overwhelm server with requests	Gaming server attacks
SQL Injection	Malicious DB query	Extracting user passwords
Man-in-the-Middle	Intercept communications	Public WiFi attacks
Social Engineering	Manipulate humans	Impersonating IT support
Zero-day exploit	Unknown vulnerability	Log4Shell (2021)

SECURITY MEASURES

Technical Defences

- **Encryption:** scrambles data (AES, RSA) — HTTPS uses TLS
- **Firewalls:** filter incoming/outgoing traffic by rules
- **MFA (Multi-Factor Auth):** password + biometric/OTP
- **Antivirus/EDR:** detects and removes malware
- **Patches/Updates:** fix known vulnerabilities
- **VPN:** encrypts traffic, masks IP address

Human & Process Defences

- Security awareness training
- Strong password policies (password managers)
- Principle of least privilege
- Incident response plans
- Regular security audits & penetration testing

DATA PRIVACY

Key Privacy Concepts

- **Personally Identifiable Information (PII):** name, email, location, ID numbers
- **Data minimisation:** only collect what is necessary
- **Anonymisation:** remove identifying info from data
- **Pseudonymisation:** replace ID with alias
- **Consent:** users must agree to data collection
- **Right to erasure:** users can request data deletion ("right to be forgotten")

GDPR (General Data Protection Regulation)

- EU law enforced since 2018
- Applies to any org handling EU citizens' data
- Key rights: access, rectification, erasure, portability
- Fines up to 4% of global annual revenue
- Requires explicit consent, data breach notification

⚖ **EVALUATION** **Privacy vs Security Tension:** Governments argue mass surveillance is necessary for national security (e.g., NSA PRISM). Critics argue it violates human rights, chills free expression, and lacks accountability.
Key question: Who should have oversight of surveillance powers?

💡 **EXAM TIP** Case study to know: **Edward Snowden (2013)** — leaked NSA mass surveillance documents. Raises questions about whistleblowing, state surveillance, digital rights, and the balance between security and privacy.

⚠ **COMMON MISTAKE** Don't write "encryption makes data safe". Write: "encryption reduces the risk of unauthorised access but does not eliminate all vulnerabilities — human factors and key management remain risks."

👫 S4 — Social Impacts: Digital Divide & Access

📉 THE DIGITAL DIVIDE

Definition

- Gap between those with and without access to digital technologies
- Not just about devices — includes skills, connectivity, relevant content
- Operates at global, national, and community levels

Dimensions of the Digital Divide

- **Access divide:** physical availability of devices & internet
- **Skills divide:** digital literacy, ability to use technology effectively
- **Usage divide:** meaningful vs passive use
- **Content divide:** lack of local-language or relevant digital content

Causes of the Digital Divide

- Poverty — cannot afford devices or data
- Infrastructure — no broadband in rural areas
- Education — low digital literacy
- Age — elderly less comfortable with technology
- Geography — remote/landlocked regions disadvantaged
- Language — most internet content is in English

🌍 GLOBAL ACCESS

Key Statistics Context

- ~5.4 billion internet users globally (2024)
- Significant gap between developed and developing nations
- Sub-Saharan Africa and South/Southeast Asia have lowest connectivity
- Mobile internet has helped bridge the gap (mobile-first societies)

Initiatives to Bridge the Divide

- **Meta's Free Basics:** free limited internet access — criticised for violating net neutrality
- **Google Loon (ended 2021):** balloon-based internet over remote areas
- **Starlink (SpaceX):** satellite internet, now available in 70+ countries
- **One Laptop per Child:** low-cost laptops for developing nations
- **UN Sustainable Development Goal 9:** universal internet access by 2030

⚖ EVALUATION **Critical View:** Corporate-led solutions (Meta, Google) may create dependency and serve commercial interests rather than genuine development goals. Community-owned infrastructure may be more equitable.

⚖ EVALUATION **Digital Divide Evaluation:**
Positive trend: Mobile phones have democratised access — 6+ billion mobile subscribers globally; mobile banking (M-Pesa in Kenya) has transformed financial inclusion
Persistent challenge: Meaningful access (fast broadband, skills, relevant content) remains deeply unequal — digital inequality risks reinforcing existing social inequalities
Key tension: Technology companies profit from expansion into developing markets — whose interests are truly served?

💡 EXAM TIP When discussing the digital divide, always distinguish between **access** and **meaningful use**. Owning a smartphone ≠ full digital inclusion if literacy and connectivity are poor.

👫 S4 — Social Media & Communication Changes

📱 SOCIAL MEDIA IMPACTS

Positive Impacts

- Global connectivity — maintain relationships across borders
- Amplifies marginalised voices (#MeToo, #BlackLivesMatter)
- Enables citizen journalism and rapid news sharing
- Crowdfunds charitable causes
- Educational content (YouTube tutorials, Twitter threads)
- Business opportunities — SME marketing

Negative Impacts

- **Misinformation/Disinformation:** rapid, unfiltered spread
- **Echo chambers:** algorithms reinforce existing beliefs
- **Cyberbullying:** harassment online — linked to youth mental health issues
- **Addiction:** dopamine-driven engagement loops
- **Data exploitation:** platforms monetise personal data
- **Polarisation:** extreme content gets more engagement

🔄 HOW COMMUNICATION HAS CHANGED

Aspect	Pre-Digital	Digital Age
Speed	Days (post)	Instant (messaging)
Reach	Local/national	Global
Format	Text, phone	Text, voice, video, AR
Gatekeepers	Editors, broadcasters	Algorithms, platforms
Permanence	Ephemeral or archived	Often permanent (screenshots)
Anonymity	Difficult	Easy (creates new risks)

⚖ EVALUATION **Filter Bubbles (Eli Pariser):** Algorithms show users content they already agree with → reduced exposure to diverse viewpoints → political polarisation and groupthink. Platforms profit from engagement, not truth.

📰 MISINFORMATION VS DISINFORMATION

Misinformation

- False information spread **unintentionally**
- Example: sharing inaccurate COVID health advice
- Cause: lack of media literacy

Disinformation

- False information spread **deliberately**
- Example: state-sponsored election interference
- Cause: political/commercial motive

Countering It

- Media literacy education
- Platform fact-checking labels
- Algorithm transparency
- Regulation (e.g., EU DSA)

💡 EXAM TIP Case study: **Cambridge Analytica (2018)** — Facebook data of 87 million users harvested without consent and used for political microtargeting. Links: data privacy, social media, democracy, consent, regulation.

⚠ COMMON MISTAKE Don't just say "social media is bad for mental health". Nuance is required: evidence is mixed; individual factors matter; platforms have begun adding wellbeing features. Acknowledge the debate.

👫 S4 — Economic Impacts & Future of Work

💰 ECONOMIC IMPACTS OF TECHNOLOGY

Positive Economic Effects

- New industries created (gig economy, app development)
- Global e-commerce enables small businesses to reach global markets
- Productivity gains — automation, digital tools
- Reduced transaction costs (digital payments, blockchain)
- Financial inclusion — mobile banking in developing regions
- Platform economy creates flexible work (Uber, Fiverr)

Negative Economic Effects

- Job displacement — automation eliminates routine roles
- Wealth concentration — tech monopolies (Big Tech)
- Tax avoidance — multinational tech firms exploit loopholes
- Gig economy = precarious work, no benefits
- Surveillance capitalism — user data as commodity
- Winner-takes-all markets — small players squeezed out

🤖 AUTOMATION & FUTURE OF WORK

Jobs at Risk of Automation

- Routine manual: manufacturing, assembly, transportation
- Routine cognitive: data entry, basic accounting, customer service
- **McKinsey (2017):** up to 800 million jobs may be displaced by 2030

Jobs Likely to Grow

- Creative roles — design, writing, art (AI cannot fully replicate creativity)
- Technical: AI/ML engineers, cybersecurity, data scientists
- Human-centric: social work, therapy, nursing
- Hybrid roles: "AI trainers", prompt engineers

⚖ **EVALUATION** **Optimist view:** Technology creates more jobs than it destroys (Industrial Revolution precedent); frees humans from dangerous/repetitive work
Pessimist view: This time is different — AI can replicate cognitive tasks; displacement faster than reskilling; inequality grows

🏢 CHANGING WORK PATTERNS

Remote Work

- COVID-19 accelerated adoption
- Tools: Zoom, Slack, Teams
- Pros: flexibility, reduced commute, global talent access
- Cons: isolation, blurred work-life boundaries, digital divide in access

Gig Economy

- Short-term contracts via platforms
- Examples: Uber, Deliveroo, Upwork
- Pros: flexibility, entrepreneurship
- Cons: no sick pay, no pension, algorithmic control over workers

Digital Skills Gap

- Mismatch between skills employers need and available workforce
- Solution: lifelong learning, reskilling programmes
- Policy response: UBI proposals, digital apprenticeships
- Education systems slow to adapt

💡 **EXAM TIP** Link economic impacts to specific companies for marks: mention **Amazon's warehouse automation**, **Uber's gig model**, or **Apple's supply chain in developing nations** as concrete case studies.

⚖ S5 — Governance & Ethics: Data Protection & Regulation

📜 DATA PROTECTION FRAMEWORKS

GDPR (EU, 2018)

- **Scope:** any org processing EU residents' data
- **Principles:** lawfulness, fairness, transparency, purpose limitation, data minimisation, accuracy, storage limitation, integrity & confidentiality
- **Rights:** access, rectification, erasure, restriction, portability, object
- **Fines:** up to €20M or 4% global revenue
- **Example fine:** Meta fined €1.2B (2023) for data transfers

Other Notable Frameworks

- **CCPA (California, USA):** consumer privacy rights in California
- **PDPA (India, 2023):** India's Digital Personal Data Protection Act
- **PIPL (China, 2021):** China's personal information law
- **COPPA (USA):** protects children under 13 online

🏛 REGULATION OF TECHNOLOGY

Approaches to Regulation

- **Government-led:** laws passed by legislatures (GDPR, AI Act)
- **Self-regulation:** industry sets own standards (often criticised as insufficient)
- **Co-regulation:** government + industry collaboration
- **International treaties:** cross-border coordination (rare, difficult)

EU AI Act (2024)

- World's first comprehensive AI legislation
- Risk-based approach: unacceptable → high → limited → minimal risk
- **Banned:** mass surveillance, social scoring, manipulative AI
- **High risk:** biometrics, critical infrastructure, employment AI — strict requirements
- **General purpose AI (GPAI):** must disclose training data, meet transparency rules

⚖ **EVALUATION** **Regulation tension:** Over-regulation may stifle innovation; under-regulation enables harm. Democratic societies must balance innovation ecosystems with human rights protections.

🌐 INTERNET GOVERNANCE

Key Bodies

- **ICANN:** manages domain names & IP addresses
- **ITU:** UN agency for telecom standards
- **W3C:** sets web standards (HTML, CSS)
- **IGF:** multi-stakeholder internet governance forum

Net Neutrality

- Principle: all internet traffic treated equally
- ISPs should not throttle/prioritise certain content
- EU has net neutrality law; USA repealed it (2017)
- Debate: innovation vs equality of access

Content Moderation

- Platforms must remove illegal content
- Tension: free speech vs harm prevention
- EU DSA (Digital Services Act) holds platforms accountable
- Algorithmic amplification of harmful content is contested

⚖ S5 — Ethical Issues: AI Bias, Surveillance & Rights

🤖 ETHICAL ISSUES IN AI

AI Bias

- Trained on historical data → encodes historical inequality
- **Example:** facial recognition less accurate for dark skin tones (MIT study — Joy Buolamwini)
- **Example:** Amazon hiring AI penalised female applicants
- **Example:** COMPAS risk assessment tool — racially biased in US courts
- **Fix:** diverse training data, bias audits, diverse development teams

AI Accountability

- **Black box problem:** many AI systems lack explainability
- **GDPR Article 22:** right not to be subject to purely automated decisions
- **Explainable AI (XAI):** growing field — making AI decisions interpretable
- **Liability gap:** if an AI makes a harmful decision, who is responsible? Developer? Deployer? User?

👁 SURVEILLANCE & DIGITAL RIGHTS

State Surveillance

- CCTV networks — UK has densest CCTV per capita
- China's Social Credit System — scores citizens on behaviour
- Mass metadata collection — phone/email data harvesting
- **Snowden revelations:** NSA collected data on millions globally
- **Argument for:** national security, crime prevention
- **Argument against:** chilling effect on speech, human rights violations

Corporate Surveillance

- Platforms track behaviour for targeted advertising
- **Surveillance capitalism (Shoshana Zuboff):** human experience as free raw material for commercial prediction products
- Examples: Google, Facebook, TikTok data practices

⚖ KEY ETHICAL FRAMEWORKS FOR IB EVALUATION

Utilitarianism

- Greatest good for greatest number
- Supports: mass data collection if it benefits majority
- Criticised: ignores individual rights of minorities

Deontology

- Actions are right/wrong regardless of outcomes
- Privacy is a right — surveillance is wrong even if "effective"
- Supports: strong data protection laws

Virtue Ethics

- Focus on character & intentions
- Should tech companies be "virtuous" actors?
- Corporate responsibility & values-driven design

⚖ **EVALUATION Key Evaluation: Right to Privacy vs Public Safety**

The tension between surveillance for security and the right to privacy is a core IB theme. Strong answers will: (1) acknowledge both sides with evidence, (2) distinguish between proportionate vs disproportionate surveillance, (3) discuss accountability mechanisms (judicial oversight, transparency reports), and (4) reach a reasoned conclusion about where the boundary should lie.

🔴 HL Extensions — Advanced Digital Systems Analysis

HL ONLY

The following content extends core SL topics with greater depth, complexity, and critical analysis as expected at Higher Level.

🏗 ADVANCED SYSTEMS ARCHITECTURE

HL

Von Neumann Architecture

- Single shared memory for instructions & data
- Bottleneck: CPU and memory share same bus (Von Neumann bottleneck)
- Alternative: Harvard Architecture — separate memory buses
- Modern CPUs use cache to partially overcome bottleneck

HL

Parallel Processing

- **Multi-core CPUs:** multiple processors on one chip
- **GPU computing:** thousands of small cores for parallel tasks (AI training)
- **Distributed computing:** multiple networked machines work together (cloud HPC)
- Critical for: AI training, weather modelling, genomics

HL

Virtualisation

- One physical machine runs multiple virtual machines (VMs)
- Hypervisor manages virtual environments
- Enables: cloud computing, efficient server use, sandboxing
- Containers (Docker) = lightweight alternative to VMs

🧪 ADVANCED AI ANALYSIS

HL

Large Language Models (LLMs)

- Trained on billions of text tokens via transformer architecture
- Examples: GPT-4, Claude, Gemini, LLaMA
- Capabilities: text generation, code, reasoning, translation
- **Hallucination:** generates plausible but false information
- **RLHF:** Reinforcement Learning from Human Feedback — aligns outputs
- **Emergent behaviour:** unexpected capabilities appear at scale

HL

Ethical AI Design Principles

- **Fairness:** equal performance across demographic groups
- **Accountability:** clear responsibility for AI decisions
- **Transparency:** explainable reasoning
- **Privacy:** data minimisation, federated learning
- **Human oversight:** "human in the loop" for high-stakes decisions
- **Safety:** alignment research — ensuring AI acts within intended bounds

HL

Blockchain & Decentralisation

- Distributed ledger — records stored across many nodes
- Immutable — cannot be altered without consensus
- **Uses:** cryptocurrency, smart contracts, supply chain, NFTs
- **PoW vs PoS:** Proof of Work (energy-intensive) vs Proof of Stake (efficient)
- **Critique:** speculation, fraud, environmental impact

🔴 HL Extensions — Deep Case Studies & Complex Debates

HL ONLY

📰 HL CASE STUDY: CHINA'S DIGITAL ECOSYSTEM

HL

The Great Firewall

- China's internet censorship system
- Blocks: Google, Facebook, YouTube, WhatsApp, Wikipedia
- Uses: IP blocking, DNS poisoning, deep packet inspection
- Domestic alternatives: WeChat, Baidu, Weibo, Alibaba

HL

Social Credit System

- Government scores citizens based on financial, social, legal behaviour
- Consequences: travel bans, school access denied for low scorers
- **Critique:** oppressive, enables authoritarian control
- **Chinese gov view:** promotes social trust, reduces corruption
- Raises: fundamental questions about state power in digital age

HL

Digital Sovereignty

- Nations asserting control over their digital infrastructure & data
- EU: "tech sovereignty" — reducing dependence on US/China platforms
- Russia: Runet — attempt at isolated national internet
- Tension: global internet vs national control

💬 HL COMPLEX ETHICAL DEBATES

HL

AI Existential Risk

- Debate among researchers: does advanced AI pose existential threats?
- **Concerned:** Stuart Russell, Geoffrey Hinton — misaligned superintelligent AI
- **Sceptical:** Yann LeCun — current AI far from general intelligence
- Policy responses: AI Safety summits (Bletchley Park, 2023)

HL

Intellectual Property in the Digital Age

- Copyright in digital content — easy to copy and share
- **AI and copyright:** who owns AI-generated content?
- Training AI on copyrighted data — lawsuits against OpenAI, Meta
- Creative Commons vs traditional copyright models
- Open source vs proprietary AI models

HL

Deepfakes & Synthetic Media

- AI-generated fake images, video, audio of real people
- **Harms:** non-consensual intimate imagery, political manipulation, fraud
- **Detection challenge:** deepfake detection arms race
- **Regulation:** EU AI Act classifies deepfakes — must be labelled
- Example: deepfake audio of political leaders used in elections

⚖ EVALUATION **HL Extended Evaluation: Should AI development be internationally regulated?**
For: AI risks cross borders; unregulated development creates race-to-bottom; democratic alignment needed
Against: Geopolitical competition makes agreement unlikely; different values between nations (China vs EU vs USA); regulation may disadvantage some states

> **Conclusion:** A tiered approach — domestic strong regulation + international norms for frontier AI — is most realistic

✕ Exam Mastery — Common Mistakes to Avoid

⚠ TOP 15 STUDENT MISTAKES IN IB DIGITAL SOCIETY

⚠ COMMON MISTAKE **Mistake 1:** Describing instead of evaluating

IB wants analysis, not description. "Social media has many effects on society" earns no marks. Evaluate: argue, justify, conclude.

⚠ COMMON MISTAKE **Mistake 2:** One-sided arguments

Never argue only pros or only cons. IB expects balanced perspectives with a reasoned conclusion.

⚠ COMMON MISTAKE **Mistake 3:** No real-world examples

"AI is used in healthcare" scores low. "DeepMind's AI detected diabetic retinopathy in NHS trials" earns marks.

⚠ COMMON MISTAKE **Mistake 4:** Confusing data with information

Data = raw facts. Information = processed data with context. Use both terms precisely.

⚠ COMMON MISTAKE **Mistake 5:** Ignoring stakeholder perspectives

Always consider multiple stakeholders: individuals, businesses, governments, developers, marginalised groups.

⚠ COMMON MISTAKE **Mistake 6:** Treating correlation as causation

Two things happening together does not mean one causes the other. State "associated with" not "causes".

⚠ COMMON MISTAKE **Mistake 7:** Confusing WWW and the internet

The internet is the infrastructure. WWW is one service running on it. They are NOT synonymous.

⚠ COMMON MISTAKE **Mistake 8:** Overlooking context in ethics questions

Context matters. CCTV in a high-crime area is different from CCTV in a democracy surveilling protest movements.

⚠ COMMON MISTAKE **Mistake 9:** Not defining key terms

Define every major term you use (AI, cloud computing, Big Data). This shows precision and earns marks.

⚠ COMMON MISTAKE **Mistake 10:** Ignoring the digital divide's complexity

Don't reduce to "poor countries don't have internet." Address skills divide, content relevance, mobile-first realities.

⚠ COMMON MISTAKE **Mistake 11:** Saying "AI thinks/knows/understands"

AI does not think or understand. It pattern-matches. Use precise language: "the model outputs", "the algorithm classifies".

⚠ COMMON MISTAKE **Mistake 12:** Not answering the command term

"Discuss" ≠ "Describe". Check Page 18 for what each command term requires and structure accordingly.

⚠ COMMON MISTAKE **Mistake 13:** Claiming technology is always positive or negative

Technology is a tool. Its impact depends on design, governance, access, and use context. Avoid technological determinism.

⚠ COMMON MISTAKE **Mistake 14:** No conclusion in evaluation questions

Always end with a justified conclusion. "Overall, while X has merits, Y is more significant because..."

⚠ COMMON MISTAKE **Mistake 15:** Forgetting HL depth requirements

(HL) HL responses must show greater critical depth, more complex analysis, and more sophisticated evaluation than SL.

🎯 Exam Strategy — Structure & Command Terms

📋 IB COMMAND TERMS — KNOW THESE

Define
Give the precise meaning of a word or concept. No examples needed unless asked.

Describe
Give a detailed account. State what it is, how it works, what it looks like.

Outline
Give a brief account or summary. Less detail than "describe".

Explain
Give a detailed account with reasons or causes. Show understanding.

Analyse
Break into components and show how they relate. Go beyond surface.

Discuss
Offer a balanced review of multiple viewpoints, evidence, and arguments.

Evaluate
Make a judgement based on evidence. Weigh strengths and limitations. Conclude.

Examine
Consider an argument carefully; query and appraise its relevance and validity.

Compare
Give an account of similarities and differences.

📝 ANSWER STRUCTURE: SHORT RESPONSE

For "Explain" / "Describe" (4–6 marks)

- **Point:** State the concept clearly
- **Explanation:** Why/how does it work?
- **Example:** Real-world illustration
- **Link:** Connect back to the question

"Cloud computing allows users to store data on remote servers [P], which means files can be accessed from any device with an internet connection [E]. For example, Google Drive enables students to collaborate on documents from different locations [Ex]. This reduces dependency on local hardware [L]."

📝 ANSWER STRUCTURE: EVALUATE / DISCUSS (8–12 MARKS)

The PEEL-C Structure

Point — State one side of the argument

Evidence — Real example, statistic, case study

Explain — Why does this support your point?

Link — Relate back to question

→ Repeat for opposite perspective

Conclusion — Weigh arguments, state justified verdict

⏱ TIME MANAGEMENT IN THE EXAM

Paper 1 Strategy

- Read all questions first — choose wisely
- 1 mark ≈ 1 key point
- Underline command terms before writing
- Don't write essays for 2-mark questions

Paper 2 Strategy

- Plan 5 minutes before writing
- Use subheadings to structure long answers
- Always use real-world examples
- Save 3 mins to re-read & check conclusion

General Tips

- Write legibly — examiners cannot mark what they can't read
- Answer every question — no penalty for wrong answers
- Refer back to the case study when given
- Use IB terminology throughout

📖 Key Terms — Essential Definitions

🖥️ DIGITAL SYSTEMS

Algorithm	A step-by-step set of instructions for solving a problem or performing a computation
Binary	Base-2 number system using only 0 and 1; fundamental to all digital data
CPU	Central Processing Unit — the brain of a computer; executes instructions
Embedded system	A dedicated computer within a physical device with a specific function
Firmware	Software permanently stored in hardware (e.g., BIOS); controls hardware startup
Operating system	Software managing hardware resources and providing services for applications
Virtualisation	Technology creating virtual versions of hardware, enabling multiple VMs on one physical machine

📊 DATA & AI

Big Data	Extremely large datasets characterised by volume, velocity, variety, veracity, and value
Machine learning	AI systems that learn from data without being explicitly programmed for each task
Neural network	An AI model with layers of interconnected nodes inspired by the human brain
Algorithmic bias	Systematic discrimination in AI outputs resulting from biased training data or design
Hallucination	When an AI model generates plausible-sounding but factually incorrect information
Metadata	Data about data — e.g., timestamps, file sizes, sender/receiver info

🌐 NETWORKS & SECURITY

Bandwidth	Maximum data transfer rate of a network connection (bits per second)
DNS	Domain Name System — translates human-readable domain names to IP addresses
Encryption	Converting data into a coded format to prevent unauthorised access
Firewall	Security system that monitors and controls network traffic based on predefined rules
Protocol	A set of rules governing how data is transmitted and received across a network
VPN	Virtual Private Network — encrypts internet traffic and masks IP address
Phishing	Fraudulent attempt to obtain sensitive info by impersonating a trustworthy entity

⚖️ SOCIAL & ETHICS

Digital divide	Gap between those with and without meaningful access to digital technologies
Filter bubble	Algorithmic curation that limits users' exposure to diverse viewpoints
GDPR	EU General Data Protection Regulation — comprehensive data protection law (2018)
Net neutrality	Principle that all internet traffic should be treated equally by ISPs
Surveillance capitalism	Business model where user data is collected and sold to predict and influence behaviour
Digital sovereignty	A nation's ability to control its own digital infrastructure and data flows
Disinformation	Deliberately false information spread to deceive; contrasted with unintentional misinformation

🌙 Final Revision — Night Before Exam Checklist

✅ NIGHT BEFORE CHECKLIST

- ☐ I can explain the input-process-output model with examples
- ☐ I can distinguish hardware from software with examples
- ☐ I can explain cloud computing (IaaS, PaaS, SaaS) and evaluate it
- ☐ I can explain AI, ML, and neural networks clearly
- ☐ I can describe IoT and give real-world examples
- ☐ I can define data vs information and the 5 V's of Big Data
- ☐ I can identify and explain algorithmic bias with a real case
- ☐ I can describe at least 4 network types (PAN, LAN, WAN, VPN)
- ☐ I can explain DNS, TCP/IP, and HTTPS in simple terms
- ☐ I can describe 5 cybersecurity threats and corresponding defences
- ☐ I can explain GDPR key principles and rights
- ☐ I can evaluate the digital divide from multiple perspectives
- ☐ I can evaluate social media's impact on society
- ☐ I can discuss automation and future of work with both sides
- ☐ I know the Cambridge Analytica and Snowden case studies
- ☐ I understand all command terms and can apply them correctly
- ☐ I know the PEEL-C structure for extended responses

⚡ RAPID SUMMARY — KEY FACTS

- AI types: Narrow (real), General (theoretical), Super (hypothetical)
- Cloud services: IaaS = infrastructure, PaaS = platform, SaaS = software
- Big Data 5 V's: Volume, Velocity, Variety, Veracity, Value
- Internet layers: user → router → ISP → backbone → server
- GDPR fines: up to €20M or 4% global revenue (whichever higher)
- HTTPS = HTTP + TLS encryption
- Digital divide: access + skills + usage + content gaps
- Filter bubble = algorithmic echo chamber (Eli Pariser, 2011)
- Surveillance capitalism = Shoshana Zuboff's concept
- EU AI Act (2024): world's first comprehensive AI law
- Cambridge Analytica: 87M Facebook profiles, political targeting
- Snowden: revealed NSA's PRISM mass surveillance (2013)
- Von Neumann: shared memory bus → potential bottleneck (HL)
- COMPAS: racially biased recidivism AI used in US courts
- M-Pesa: mobile banking in Kenya — digital financial inclusion

🌄 EXAM MORNING MINDSET

> 💡 **EXAM TIP** **Before the exam:** Read each question fully before writing. Underline the command term. Plan extended responses in 60 seconds. Always include a named real-world example. End evaluate/discuss with a reasoned conclusion. You know this — go show them.

IB DP Digital Society

Complete Cheat Sheet · All Topics Covered

Good luck — you've got this 🚀